

Three Frameworks You Can Use to Develop a Hybrid Mobile App

Creating a mobile app isn't as difficult or as complicated as it sounds. The tools are all available on the Internet. With a little bit of the right guidance, your dream mobile app can be yours to share with the world. In this article, the author explains how to use Angular JS, jQuery Mobile and PhoneGap, and then goes on to giving a step-by-step outline on how to develop a hybrid mobile app.



Let's make a hybrid mobile app using Angular JS, jQuery Mobile and PhoneGap. Before starting with the code, I would like to explain a little about what a hybrid mobile app is, and what Angular JS, jQuery Mobile and PhoneGap are.

Mobile apps are generally of three major types—native apps, mobile Web apps and hybrid mobile apps. Native apps are made using the platform-specific SDK, tools and languages provided by the platform vendor. So, for instance, you can make native Android apps using Java with the Android SDK. Mobile Web apps, on the other hand, are websites or Web applications optimised for mobile browsers. These are developed using Web development technologies - HTML, CSS and JavaScript, but they do not have the luxury of accessing the native features of the mobile. Hybrid mobile apps use the best of both the worlds, and are made using Web development technologies like HTML, CSS and JavaScript, but they run inside a native container utilising the device's browser engine to render the HTML and process JavaScript. Beyond this, there are abstraction layers available that enable developers to access the native features of the device as well.

jQuery Mobile is a very famous touch-optimised Web/mobile framework built on top of the jQuery core, by the jQuery team. It has a very minimal learning curve. It is robust enough and the views created using jQuery Mobile adjust themselves automatically based on the browser's form factor. In other words, there is a very minimalistic CSS change that

the developer needs to do when it comes to designing views for various form factors using jQuery Mobile.

Angular JS is a MVW framework that focuses on extending traditional HTML to cater to dynamic content. MVW stands for Model-View-Whatever, where 'whatever' implies "whatever works for you", though some people also like to bucket Angular JS as a MVC or a MV* framework. Angular JS stands out from other frameworks because it eliminates DOM manipulation and ensures automatic synchronisation between models and views.

PhoneGap is an open source framework that helps in building hybrid mobile apps (cross-platform apps). It helps to wrap the HTML/CSS/JavaScript code inside the native container for the respective target platform, and also provides an abstraction layer in the form of a JavaScript library to access native device features.

Having discussed each of the technologies to be used in building our app, let us now jump into some coding. Let's develop a single-view app where the user will be able to fetch a list of all the car companies from around the world, then select one company from the list, and find out all the models that belong to that company.

The app structure will be as follows. At the root level,